



COMPARIS ON OF IP ROUTING PROTOCOL S

RIPv2, OSPF & EIGRP

Networking Concepts

Title of Assignment: Comparison of IP Routing Protocols

Module: CI5210

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The labelling of titles refers to Appendix A, where a screenshot of the coursework brief is provided. The numbering of the titles is directly linked with the coursework requirements to make marking simpler for the examiner.

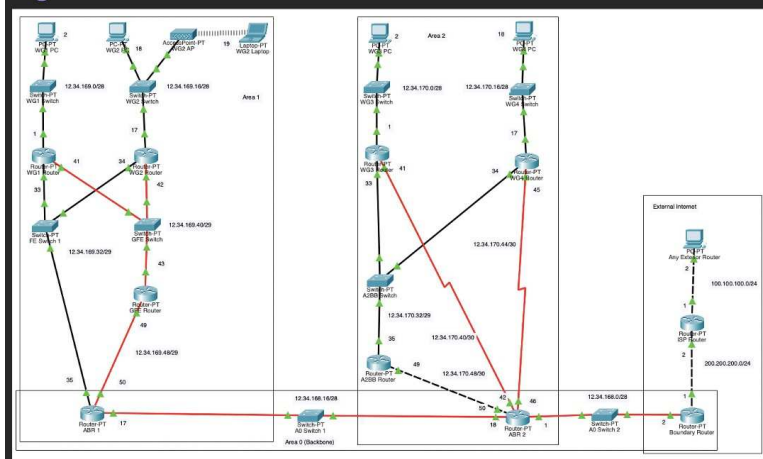
This coursework has been submitted by **Malik Omar (Student ID: K2335386)** as part of the **CI5210 Networking Concepts** coursework at Kingston University. The project involved the configuration and testing of RIPv2 and OSPF protocols within a structured topology in Cisco Packet Tracer.

7.1 Introduction

What are Routing Protocols & How will we use them?

A routing protocol is a set of guidelines that routers use to decide how best to forward data through networks that are connected. The fault tolerance, capacity, and adaptability that define modern internetworks depend on these protocols. Routing protocols enable routers to constantly learn and adapt to topology changes instead of depending on static, manually specified routes.

Figure 1:



This report covers three important routing protocols: OSPF, a link-state protocol; EIGRP, a hybrid protocol; and RIPv2, a distance-vector protocol. Cisco Packet Tracer was used to configure and test each in a fault-tolerant network design. Through simulation and analysis, this study illustrates their variations in behaviour, efficiency, route computation, and convergence performance.

(The complete internetwork architecture used to develop and compare the protocols is shown in Figure 1.)

7.2 Theory of Routing Protocols

Routers depend on routing protocols to acquire information about routing paths and determine the most efficient ways to forward data packets across networks. These protocols monitor network topology while dynamically updating routing tables to reflect changes which results in an efficient network operation with fault tolerance capabilities (via dynamic routing and fault tolerance features). Routing protocols can be divided into three major groups: distance-vector, link-state, and hybrid protocols, as they rely on different algorithms and mechanisms for path selection and convergence. [3]

RIPV2 - DISTANCE VECTOR PROTOCOL

RIPv2 (Routing Information Protocol version 2) is a distance vector protocol utilising the Bellman-Ford algorithm [7] to establish the optimal route based on the number of hops. Every 30 seconds, each router sends its full routing table to its directly connected neighbours, making it best suited for small, static networks as this periodic update increases bandwidth usage in larger networks. The maximum number of hops is 15, and if a destination is at 16 hops or more, it is unreachable.

RIPv2 supports CIDR [1] (Classless Inter-Domain Routing) and uses multicast (224.0.0.9) [2] for updates. Although it is simple to configure, RIPv2 converges slowly and does not scale well. To

mitigate routing loops [3], RIP implements:

Why RIP is used [3]:

1. Split horizon - prevents routing information from being sent back on the same interface it was received.
2. Route poisoning - marks failed routes with a metric of 16 (infinity).
3. Hold-down timers - prevent rapid re-advertisement of invalid routes.

Therefore, as previously mentioned, this makes RIP best suited for small, static networks with low complexity and topology function.

OSPF - LINK-STATE PROTOCOL

When multiple links are connecting to the same destination, OSPF determines the best path based on the OSPF cost using Dijkstra's algorithm [8]. Each router starts by exchanging Link State Advertisements (LSAs) with their neighbours to build a Link State Database (LSDB). This database is then used by all routers on the network this database to calculate a Shortest Path Tree (SPT) in an optimal routing scenario. OSPF can achieve a hierarchical network design with an Area which supports Area 0 (the back bone) and Areas 1 and 2 connected via Area Border Routers (ABR) in Figure 1.

Why OSPF is used:

- Fast convergence
- More accurate path selection
- Scalability via its hierarchical model

Though more complex to set up than RIP, it is ideal for enterprise-scale networks.

EIGRP - HYBRID PROTOCOL

Open Shortest Path First, or OSPF The Cisco-proprietary hybrid protocol known as EIGRP (Enhanced Interior Gateway Routing Protocol) combines elements of the link-state and distance vector models. It provides realistic successors as backup routes and computes loop-free routes using the DUAL (Diffusing Update Algorithm) [5].

EIGRP uses a composite metric based on [4]:

- Bandwidth - contributes most heavily to the metric; lower bandwidth = higher metric. [4]
- Delay - adds up cumulative delay across all interfaces to affect path cost. [4]
- Reliability - (optional) only considered if explicitly enabled; lower reliability = higher metric [4]
- Load - (optional) represents current traffic; fluctuates often, so it's rarely used. [4]
- MTU - doesn't affect metric calculation but is used in route selection. [4]

EIGRP employs its own transport layer, RTP (Reliable Transport Protocol), to carry messages and only sends limited updates, in contrast to RIP. EIGRP is more efficient than RIP, converges more quickly, and scales effectively [6], but it can only be used in Cisco environments unless the more recent open-standard version is being used.

7.3 Configuration and Verification

The configuration, testing, and validation of the network's three dynamic routing protocols RIPv2, OSPF, and EIGRP are described in this section. In order to guarantee isolated testing under uniform topological conditions, each protocol was configured using a specific Packet Tracer file. According to the module brief, the objective was to verify successful protocol behaviour, appropriate routing table population, and internal and external connectivity.

Each subsection contains:

- A configuration summary highlighting key router setups (with figure references)
- A verification summary using show ip route, ping, and tracert to confirm route learning and network reachability
- A brief conclusion highlighting how the protocol met the routing requirements

At the end, a shutdown test summary compares the failover resilience of all three protocols under simulated link failure.

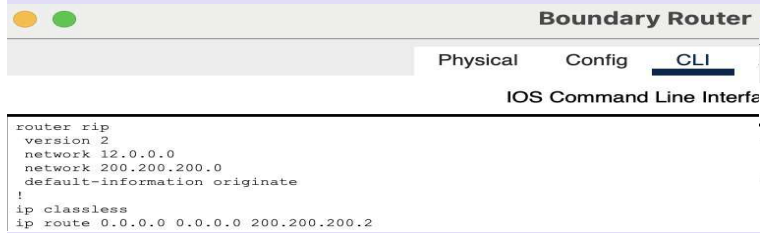
RIPV2 - HYBRID PROTOCOL

Configuration

All routers within the RIP domain (excluding the ISP router) were configured with RIPv2, and all directly connected networks were advertised. The no auto-summary command was applied to ensure correct handling of VLSM subnets and avoid automatic summarisation. The Boundary Router was configured with a static default route pointing to the ISP (0.0.0.0 0.0.0.0 200.200.200.2), and used the default-information originate command to propagate this route throughout the RIP domain. Internal routers such as WG1 advertised the networks 12.34.169.0, 12.34.169.32, and 12.34.169.40 using RIP version 2. Configuration examples are shown in Figure 2 (Boundary Router) and Figure 3 (WG1 Router), with command-line inputs detailed in Figures 4 and 5.

RIP V2 CONFIGURATION SUMMARY

FIGURE 2

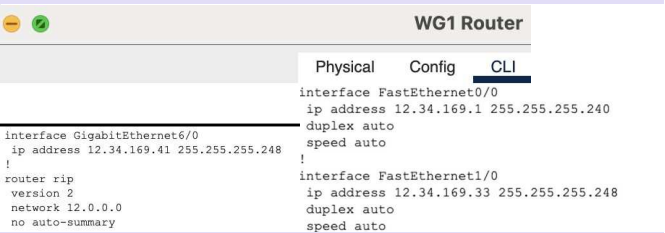


```

Boundary Router
Physical Config CLI
IOS Command Line Interface
router rip
version 2
network 12.0.0.0
network 200.200.200.0
default-information originate
!
ip classless
ip route 0.0.0.0 0.0.0.0 200.200.200.2

```

FIGURE 3

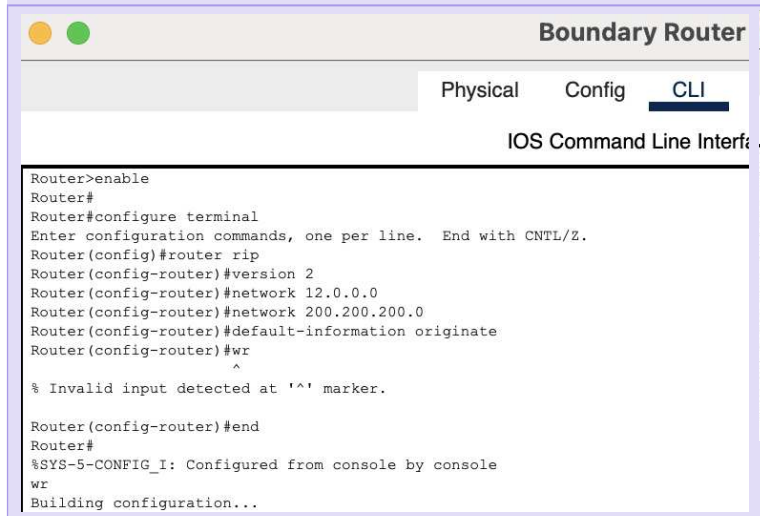


```

WG1 Router
Physical Config CLI
IOS Command Line Interface
interface FastEthernet0/0
ip address 12.34.169.1 255.255.255.240
duplex auto
speed auto
!
interface FastEthernet1/0
ip address 12.34.169.33 255.255.255.248
duplex auto
speed auto
!
interface GigabitEthernet6/0
ip address 12.34.169.41 255.255.255.248
!
router rip
version 2
network 12.0.0.0
no auto-summary

```

FIGURE 4



```

Boundary Router
Physical Config CLI
IOS Command Line Interface
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 12.0.0.0
Router(config-router)#network 200.200.200.0
Router(config-router)#default-information originate
Router(config-router)#wr
^
% Invalid input detected at '^' marker.
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...

```

FIGURE 5



```

WG1 Router
Physical Config CLI
IOS Command Line Interface
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 12.34.169.0
Router(config-router)#network 12.34.169.40
Router(config-router)#network 12.34.169.32
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]

```

Verification

The show ip route output on routers WG2, WG3, and ABR1 showed multiple entries marked with an R, indicating that routes were learned through the RIP protocol. These results, shown in Figures 6, 7, and 8, confirmed that routing information was correctly propagated throughout the RIP domain.

Connectivity was tested using ping between internal and external devices. WG1 PC successfully pinged WG3 PC (Figure 9), verifying end-to-end communication within the internal RIP-enabled network. In addition, both WG1 PC and WG2 PC successfully pinged the external server at 100.100.100.2 (Figures 10 and 11), confirming that the static default route configured on the Boundary Router was successfully learned and utilised by internal hosts.

Traceroute tests (tracert) were used to inspect the route taken by packets as they travelled across the network. The trace from WG2 PC to the external server (Figure 12) confirmed that traffic followed the correct path through the internal routers and out through the Boundary Router to the ISP. A trace from WG2 PC to WG3 PC (Figure 13) verified correct internal routing between RIP areas. Finally, a trace from WG4 PC to the external server (Figure 14) demonstrated that even remote devices within the network were able to resolve external paths correctly via the RIP-learned default route.

Overall, these results confirm that RIPv2 was correctly implemented. The protocol dynamically distributed routing information, supported reliable internal communication, and enabled full external access through the advertised static default route.

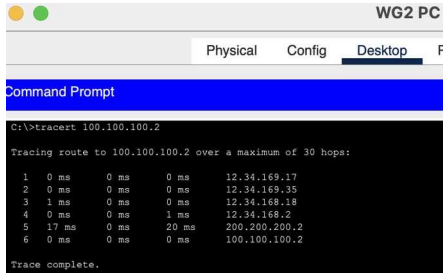
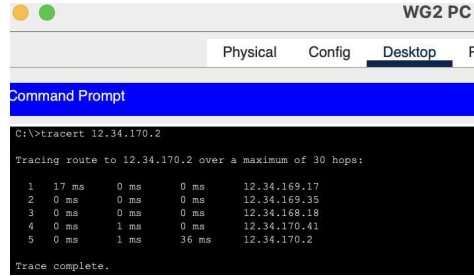
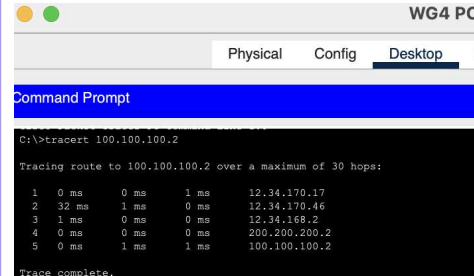
RIP V2 VERIFICATION SUMMARY			
COMMANDS		VERIFICATION TOOLS	
- show ip route (WG2, WG3, ABR1 IP Addresses) - ping (destination ip address)	SHOW IP ROUTE	PING & TRACERT	LINK SHUTDOWN TEST
	To check R* routes and learned subnets	Between PCs and the external server	To confirm RIP can reroute after failure
SHOW IP ROUTE			
WG2 ROUTER (AREA 1)		WG3 ROUTER (AREA 2)	
ABR1 (AREA 0)			
WG2 Router		WG3 Router	
ABR 1			
Physical Config CLI		Physical Config CLI	
IOS Command Line Interface		IOS Command Line Interface	
<pre>Router>show ip route Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2 E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area * - candidate default, U - per-user static route, o - ODR P - periodic downloaded static route Gateway of last resort is 12.34.169.35 to network 0.0.0.0 12.0.0.0/8 is variably subnetted, 13 subnets, 3 masks R 12.34.168.0/28 [120/2] via 12.34.169.35, 00:00:02, FastEthernet1/0 R 12.34.168.16/28 [120/1] via 12.34.169.35, 00:00:02, FastEthernet1/0 R 12.34.169.0/28 [120/1] via 12.34.169.35, 00:00:08, FastEthernet1/0 C 12.34.169.0/28 [120/1] via 12.34.169.41, 00:00:08, GigabitEthernet6/0 C 12.34.169.16/28 [120/3] via 12.34.169.35, 00:00:02, FastEthernet1/0 C 12.34.169.32/29 is directly connected, FastEthernet1/0 C 12.34.169.40/29 is directly connected, GigabitEthernet6/0 R 12.34.169.48/29 [120/1] via 12.34.169.35, 00:00:02, FastEthernet1/0 R 12.34.169.48/29 [120/1] via 12.34.169.43, 00:00:05, GigabitEthernet6/0 R 12.34.170.0/28 [120/3] via 12.34.169.35, 00:00:02, FastEthernet1/0 R 12.34.170.16/28 [120/3] via 12.34.169.35, 00:00:02, FastEthernet1/0 R 12.34.170.32/29 [120/3] via 12.34.169.35, 00:00:02, FastEthernet1/0 R 12.34.170.40/30 [120/2] via 12.34.169.35, 00:00:02, FastEthernet1/0 R 12.34.170.44/30 [120/2] via 12.34.169.35, 00:00:02, FastEthernet1/0 R 12.34.170.48/30 [120/2] via 12.34.169.35, 00:00:02, FastEthernet1/0 R 200.200.200.0/24 [120/3] via 12.34.169.35, 00:00:02, FastEthernet1/0 R* 0.0.0.0/0 [120/3] via 12.34.169.35, 00:00:02, FastEthernet1/0</pre>		<pre>Router>show ip route Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2 E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area * - candidate default, U - per-user static route, o - ODR P - periodic downloaded static route Gateway of last resort is 12.34.170.42 to network 0.0.0.0 12.0.0.0/8 is variably subnetted, 13 subnets, 3 masks R 12.34.168.0/28 [120/1] via 12.34.170.42, 00:00:27, Serial2/0 R 12.34.168.16/28 [120/1] via 12.34.170.42, 00:00:27, Serial2/0 R 12.34.169.0/28 [120/3] via 12.34.170.42, 00:00:27, Serial2/0 R 12.34.169.16/28 [120/3] via 12.34.170.42, 00:00:27, Serial2/0 R 12.34.169.32/29 [120/2] via 12.34.170.42, 00:00:27, Serial2/0 R 12.34.169.40/29 [120/3] via 12.34.170.42, 00:00:27, Serial2/0 R 12.34.169.48/29 [120/2] via 12.34.170.42, 00:00:27, Serial2/0 C 12.34.170.0/28 is directly connected, FastEthernet0/0 R 12.34.170.16/28 [120/1] via 12.34.170.34, 00:00:05, FastEthernet1/0 C 12.34.170.32/29 is directly connected, FastEthernet1/0 C 12.34.170.40/30 is directly connected, Serial2/0 R 12.34.170.44/30 [120/1] via 12.34.170.34, 00:00:05, FastEthernet1/0 R 12.34.170.48/30 [120/1] via 12.34.170.35, 00:00:16, FastEthernet1/0 R 12.34.170.48/30 [120/1] via 12.34.170.42, 00:00:27, Serial2/0 R 200.200.200.0/24 [120/2] via 12.34.170.42, 00:00:27, Serial2/0 R* 0.0.0.0/0 [120/2] via 12.34.170.42, 00:00:27, Serial2/0</pre>	
		<pre>Router>show ip route Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2 E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area * - candidate default, U - per-user static route, o - ODR P - periodic downloaded static route Gateway of last resort is 12.34.168.18 to network 0.0.0.0 12.0.0.0/8 is variably subnetted, 13 subnets, 3 masks R 12.34.168.0/28 [120/1] via 12.34.168.18, 00:00:19, FastEthernet4/0 C 12.34.168.16/28 is directly connected, FastEthernet4/0 R 12.34.169.0/28 [120/1] via 12.34.169.35, 00:00:02, FastEthernet0/0 R 12.34.169.16/28 [120/1] via 12.34.169.34, 00:00:12, FastEthernet0/0 C 12.34.169.32/29 is directly connected, FastEthernet0/0 R 12.34.169.40/29 [120/1] via 12.34.169.33, 00:00:02, FastEthernet0/0 R 12.34.169.40/29 [120/1] via 12.34.169.49, 00:00:23, GigabitEthernet6/0 C 12.34.169.48/29 is directly connected, GigabitEthernet6/0 R 12.34.170.0/28 [120/2] via 12.34.168.18, 00:00:19, FastEthernet4/0 R 12.34.170.16/28 [120/2] via 12.34.168.18, 00:00:19, FastEthernet4/0 R 12.34.170.32/29 [120/2] via 12.34.168.18, 00:00:19, FastEthernet4/0 R 12.34.170.40/30 [120/1] via 12.34.168.18, 00:00:19, FastEthernet4/0 R 12.34.170.44/30 [120/1] via 12.34.168.18, 00:00:19, FastEthernet4/0 R 12.34.170.48/30 [120/1] via 12.34.168.18, 00:00:19, FastEthernet4/0 R 200.200.200.0/24 [120/2] via 12.34.168.18, 00:00:19, FastEthernet4/0 R* 0.0.0.0/0 [120/2] via 12.34.168.18, 00:00:19, FastEthernet4/0</pre>	

FIGURE 6

FIGURE 7

PING		
WG1 PC → WG3 PC	WG1 PC → EXTERIOR PC	WG2 PC → EXTERIOR PC

RIP V2 VERIFICATION SUMMARY	
COMMANDS	VERIFICATION TOOLS

RIP V2 VERIFICATION SUMMARY			
COMMANDS		VERIFICATION TOOLS	
SHOW IP ROUTE			
WG2 ROUTER (AREA 1)		WG3 ROUTER (AREA 2)	ABR1 (AREA 0)
			
FIGURE 12		FIGURE 13	FIGURE 14

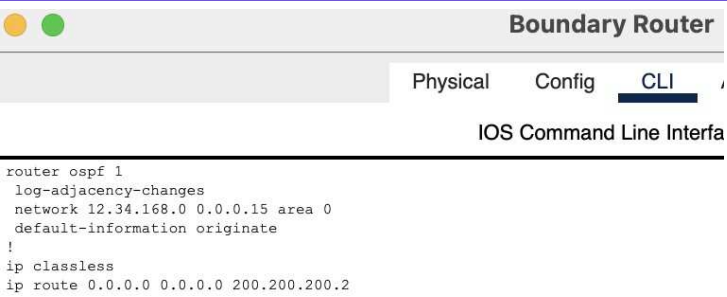
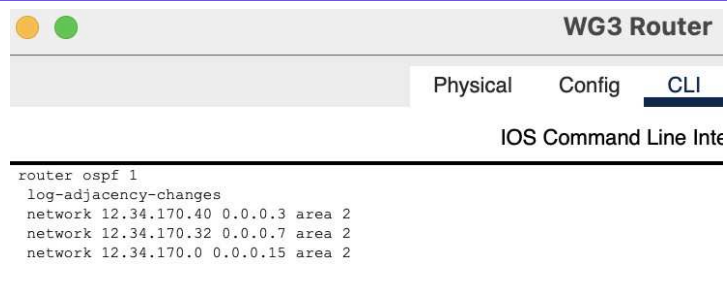
OSPF - LINK-STATE PROTOCOL

Configuration

The system had three parts, which you can see in Figure 1. ABR1 was put in the center to connect all the parts. Each router had the correct OSPF area ID set up. All the networks that were directly connected shared using the network command and the right wildcard masks.

The Boundary Router was set up to share a default route (0.0.0.0 0.0.0.0 200.200.200.2). It used a command to send this route through the OSPF network. All routers used OSPF process ID 1. The interfaces matched areas with wildcard masks. For example, the WG4 Router advertised subnets in Area 2. These subnets included 12.34.170.0, 12.34.170.16, and 12.34.170.32.

Configuration examples are shown in Figure 15 (Boundary Router) and Figure 16 (WG3 Router), with full CLI input displayed in Figures 17 and 18.

OSPF CONFIGURATION SUMMARY	
FIGURE 15	FIGURE 16
 FIGURE 17	 FIGURE 18
	

Verification

Routing tables on WG2, WG4, and ABR1 displayed O-marked entries, indicating routes learned through OSPF. These are shown in Figures 19, 20, and 21, confirming that all OSPF areas were properly interconnected and receiving routing updates.

Ping tests were then used to verify connectivity. WG1 PC successfully pinged WG4 PC (Figure 22), validating internal communication across different OSPF areas. In addition, WG1 PC and WG4 PC were both able to reach the external server at 100.100.100.2 (Figures 23 and 24), confirming that the default route from the Boundary Router was successfully propagated and used.

Traceroutes confirmed hop-by-hop routing. A trace from WG2 PC to the external server (Figure 25) followed the correct path through the OSPF domain. Further traces from WG2 PC to WG4 PC (Figure 26) and from WG4 PC to the external server (Figure 27) demonstrated that internal devices in different OSPF areas could correctly route traffic both internally and externally.

These results confirm that OSPF was correctly implemented, with area-based route propagation, inter-area communication, and stable access to the external ISP through a redistributed static default route.

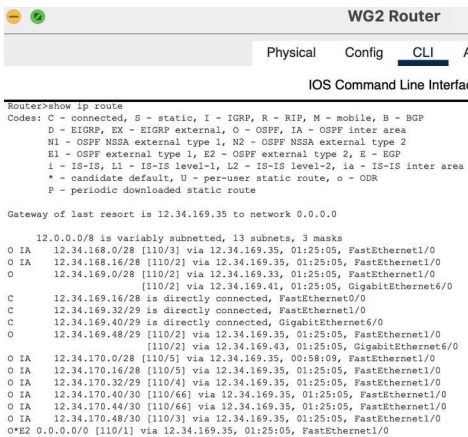
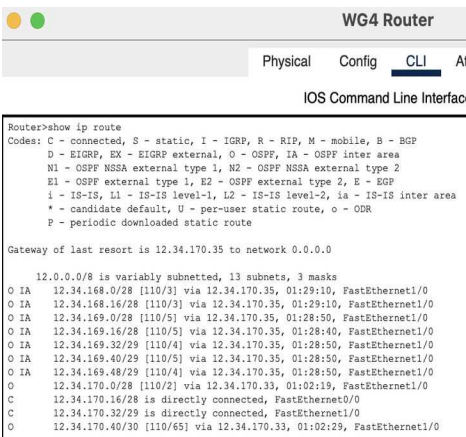
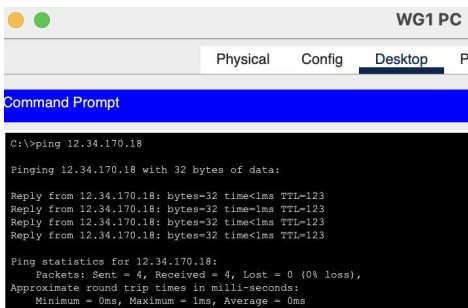
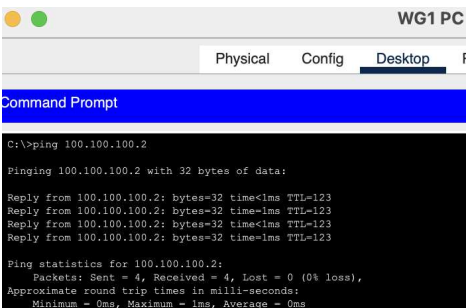
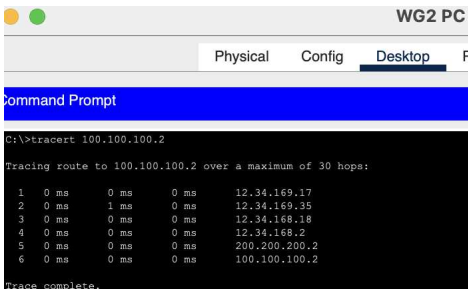
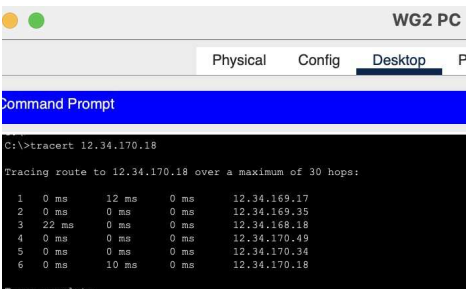
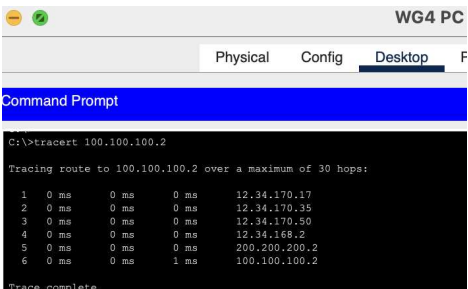
COMMANDS		VERIFICATION TOOLS	
- show ip route (WG2, WG3, ABR1 IP Addresses) - ping (destination ip address)	SHOW IP ROUTE	PING & TRACERT	LINK SHUTDOWN TEST
	To check R* routes and learned subnets	Between PCs and the external server	To confirm RIP can reroute after failure
SHOW IP ROUTE			
WG2 ROUTER (AREA 1)		WG4 ROUTER (AREA 2)	ABR1 (AREA 0)
			
FIGURE 19		FIGURE 20	
PING			
WG1 PC → WG4 PC		WG1 PC → EXTERIOR PC	WG4 PC → EXTERIOR PC
			
FIGURE 22		FIGURE 23	
TRACERT			
WG2 PC → EXTERIOR PC		WG2 PC → WG4 PC	WG4 PC → EXTERIOR PC
			
FIGURE 25		FIGURE 26	
			
		FIGURE 27	

FIGURE 28



FIGURE 29

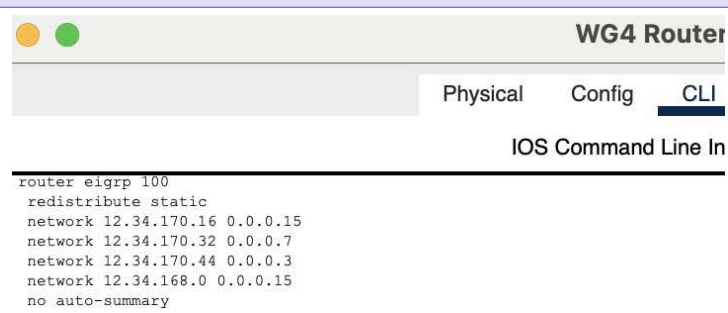


FIGURE 30

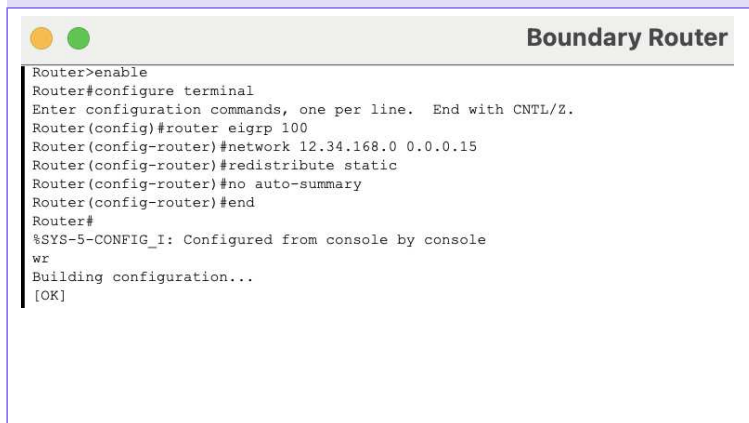
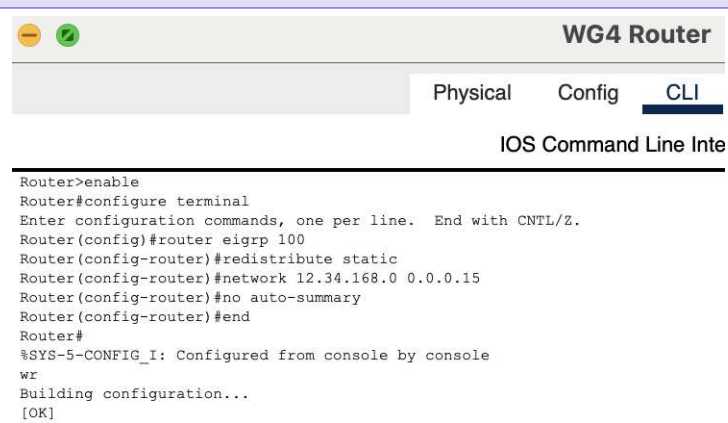


FIGURE 31



EIRGP - ADVANCED DISTANCE VECTOR PROTOCOL

Configuration

The network was divided into three areas as shown in Figure 1, with ABR1 positioned at the centre to connect all areas. Each router was configured with the correct OSPF area ID, and all directly connected networks were advertised using the network command and appropriate wildcard masks.

The Boundary Router was configured to redistribute a static default route (0.0.0.0 0.0.0.0 200.200.200.2) and used the default-information originate command to propagate this default route throughout the OSPF domain. OSPF process ID 1 was used consistently across all routers, and interfaces were matched to areas using wildcard masks. For example, WG4 Router advertised subnets in Area 2 including 12.34.170.0, 12.34.170.16, and 12.34.170.32.

Configuration examples are shown in Figure 15 (Boundary Router) and Figure 16 (WG3 Router), with full CLI input displayed in Figures 17 and 18.

Verification

In Figures 32-34 it can be seen that the Routing tables for WG2, WG3, and ABR1 entries marked with D or D* are displayed. Therefore, we know that EIGRP-learned routes were being correctly propagated. In Figure 35 connectivity is verified using ping from WG1 PC to WG3 PC and from WG1 and WG3 PCs to the external server 100.100.100.2 in Figures 36 to 37. All pings worked perfectly, showing internal devices are successfully communicating with each other and reaching the ISP via the advertised static route as intended.

Traceroutes confirmed that correct forwarding paths were established. Figure 38 shows a trace from WG2 PC to the external server showed the packet's path through internal routers to the ISP. Traces from WG2 PC to WG4 PC (Figure 39) and from WG4 PC to the external server (Figure 40) validated that inter-device communication and external access were fully operational across the EIGRP-enabled network.

Together, these results confirm that EIGRP was correctly implemented, with fast convergence, full route propagation, and end-to-end connectivity across all internal and external paths.

COMMANDS

- show ip route (WG2, WG3, ABR1 IP Addresses)
- ping (destination ip address)

VERIFICATION TOOLS

SHOW IP ROUTE

To check D* and D routes

PING & TRACERT

Between PCs and the external server

LINK SHUTDOWN TEST

To confirm EIGRP can reroute after failure

SHOW IP ROUTE

WG2 ROUTER (AREA 1)

WG2 Router			
Physical	Config	CLI	Attrib
IOS Command Line Interface			
<pre>Router>show ip route Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2 E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area * - candidate default, U - per-user static route, o - ODR P - periodic downloaded static route Gateway of last resort is 12.34.169.43 to network 0.0.0.0 12.0.0.0/8 is variably subnetted, 13 subnets, 3 masks D 12.34.168.0/28 [90/31232] via 12.34.169.43, 00:02:05, GigabitEthernet6/0 D 12.34.168.16/28 [90/28672] via 12.34.169.43, 00:02:05, GigabitEthernet6/0 D 12.34.169.0/28 [90/28416] via 12.34.169.41, 00:02:04, GigabitEthernet6/0 D 12.34.169.16/28 is directly connected, FastEthernet0/0 C 12.34.169.32/29 is directly connected, FastEthernet1/0 D 12.34.169.40/29 is directly connected, GigabitEthernet6/0 D 12.34.169.48/29 [90/3072] via 12.34.169.43, 00:02:05, GigabitEthernet6/0 D 12.34.170.0/28 [90/36352] via 12.34.169.43, 00:02:05, GigabitEthernet6/0 D 12.34.170.16/28 [90/36352] via 12.34.169.43, 00:02:05, GigabitEthernet6/0 D 12.34.170.32/29 [90/33792] via 12.34.169.43, 00:02:05, GigabitEthernet6/0 D 12.34.170.40/30 [90/20515072] via 12.34.169.43, 00:02:05, GigabitEthernet6/0 D 12.34.170.44/30 [90/20515072] via 12.34.169.43, 00:02:05, GigabitEthernet6/0 D 12.34.170.48/30 [90/31232] via 12.34.169.43, 00:02:05, GigabitEthernet6/0 D*EX 0.0.0.0/0 [170/56832] via 12.34.169.43, 00:02:05, GigabitEthernet6/0</pre>			

FIGURE 32

WG3 ROUTER (AREA 2)

WG3 Router			
Physical	Config	CLI	Attr
IOS Command Line Interface			
<pre>Router>show ip route Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2 E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area * - candidate default, U - per-user static route, o - ODR P - periodic downloaded static route Gateway of last resort is 12.34.170.35 to network 0.0.0.0 12.0.0.0/8 is variably subnetted, 13 subnets, 3 masks D 12.34.168.0/28 [90/33280] via 12.34.170.35, 00:03:34, FastEthernet1/0 D 12.34.168.16/28 [90/33280] via 12.34.170.35, 00:03:34, FastEthernet1/0 D 12.34.169.0/28 [90/36352] via 12.34.170.35, 00:03:33, FastEthernet1/0 D 12.34.169.16/28 [90/36352] via 12.34.170.35, 00:03:34, FastEthernet1/0 D 12.34.169.32/29 [90/35840] via 12.34.170.35, 00:03:34, FastEthernet1/0 D 12.34.169.40/29 [90/33792] via 12.34.170.35, 00:03:34, FastEthernet1/0 D 12.34.169.48/29 [90/33536] via 12.34.170.35, 00:03:34, FastEthernet1/0 C 12.34.170.0/28 is directly connected, FastEthernet0/0 D 12.34.170.16/28 [90/30720] via 12.34.170.34, 00:03:34, FastEthernet1/0 C 12.34.170.32/29 is directly connected, FastEthernet1/0 C 12.34.170.40/30 is directly connected, Serial1/0 D 12.34.170.44/30 [90/20514560] via 12.34.170.34, 00:03:34, FastEthernet1/0 D 12.34.170.48/30 [90/30720] via 12.34.170.35, 00:03:34, FastEthernet1/0 D*EX 0.0.0.0/0 [170/58880] via 12.34.170.35, 00:03:34, FastEthernet1/0</pre>			

FIGURE 33

ABR1 (AREA 0)

ABR 1			
Physical	Config	CLI	Attr
IOS Command Line Interface			
<pre>Router>show ip route Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2 E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area * - candidate default, U - per-user static route, o - ODR P - periodic downloaded static route Gateway of last resort is 12.34.168.18 to network 0.0.0.0 12.0.0.0/8 is variably subnetted, 13 subnets, 3 masks D 12.34.168.0/28 [90/30720] via 12.34.168.18, 00:04:39, FastEthernet4/0 C 12.34.168.16/28 is directly connected, FastEthernet4/0 D 12.34.169.0/28 [90/28672] via 12.34.169.49, 00:04:38, GigabitEthernet6/0 D 12.34.169.16/28 [90/28672] via 12.34.169.49, 00:04:39, GigabitEthernet6/0 C 12.34.169.32/29 is directly connected, FastEthernet0/0 D 12.34.169.40/29 [90/3072] via 12.34.169.49, 00:05:10, GigabitEthernet6/0 D 12.34.169.48/29 is directly connected, GigabitEthernet6/0 D 12.34.170.0/28 [90/35840] via 12.34.168.18, 00:04:39, FastEthernet4/0 D 12.34.170.16/28 [90/35840] via 12.34.168.18, 00:04:39, FastEthernet4/0 D 12.34.170.32/29 [90/33280] via 12.34.168.18, 00:04:39, FastEthernet4/0 D 12.34.170.40/30 [90/20514560] via 12.34.168.18, 00:04:39, FastEthernet4/0 D 12.34.170.44/30 [90/20514560] via 12.34.168.18, 00:04:39, FastEthernet4/0 D 12.34.170.48/30 [90/30720] via 12.34.168.18, 00:04:39, FastEthernet4/0 D*EX 0.0.0.0/0 [170/56320] via 12.34.168.18, 00:04:39, FastEthernet4/0</pre>			

FIGURE 34

PING

WG1 PC → WG3 PC

WG1 PC			
Physical	Config	Desktop	F
Command Prompt			
<pre>C:\>ping 12.34.170.2 Pinging 12.34.170.2 with 32 bytes of data: Reply from 12.34.170.2: bytes=32 time<1ms TTL=122 Reply from 12.34.170.2: bytes=32 time<1ms TTL=122 Reply from 12.34.170.2: bytes=32 time<1ms TTL=122 Reply from 12.34.170.2: bytes=32 time<1ms TTL=122 Ping statistics for 12.34.170.2: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms</pre>			

FIGURE 35

WG1 PC → EXTERIOR PC

WG1 PC			
Physical	Config	Desktop	P
Command Prompt			
<pre>C:\>ping 100.100.100.2 Pinging 100.100.100.2 with 32 bytes of data: Reply from 100.100.100.2: bytes=32 time=34ms TTL=122 Reply from 100.100.100.2: bytes=32 time<1ms TTL=122 Reply from 100.100.100.2: bytes=32 time<1ms TTL=122 Reply from 100.100.100.2: bytes=32 time<1ms TTL=122 Ping statistics for 100.100.100.2: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 34ms, Average = 8ms</pre>			

FIGURE 36

WG3 PC → EXTERIOR PC

WG3 PC			
Physical	Config	Desktop	F
Command Prompt			
<pre>C:\>ping 100.100.100.2 Pinging 100.100.100.2 with 32 bytes of data: Reply from 100.100.100.2: bytes=32 time<1ms TTL=123 Reply from 100.100.100.2: bytes=32 time<1ms TTL=123 Reply from 100.100.100.2: bytes=32 time<1ms TTL=123 Reply from 100.100.100.2: bytes=32 time=21ms TTL=123 Ping statistics for 100.100.100.2: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 21ms, Average = 5ms</pre>			

FIGURE 37

TRACERT

WG2 PC → EXTERIOR PC

WG2 PC			
Physical	Config	Desktop	F
Command Prompt			
<pre>C:\>tracert 100.100.100.2 Tracing route to 100.100.100.2 over a maximum of 30 hops: 1 0 ms 0 ms 0 ms 12.34.169.17 2 0 ms 0 ms 0 ms 12.34.169.43 3 0 ms 0 ms 0 ms 12.34.169.50 4 0 ms 0 ms 0 ms 12.34.168.18 5 0 ms 0 ms 0 ms 12.34.168.2 6 0 ms 0 ms 0 ms 200.200.200.2 7 0 ms 0 ms 0 ms 100.100.100.2 Trace complete.</pre>			

FIGURE 38

WG2 PC → WG4 PC

WG2 PC			
Physical	Config	Desktop	P
Command Prompt			
<pre>C:\>tracert 12.34.170.2 Tracing route to 12.34.170.2 over a maximum of 30 hops: 1 0 ms 0 ms 0 ms 12.34.169.17 2 0 ms 0 ms 0 ms 12.34.169.43 3 0 ms 0 ms 0 ms 12.34.169.50 4 0 ms 0 ms 0 ms 12.34.168.18 5 0 ms 0 ms 0 ms 12.34.170.49 6 0 ms 0 ms 0 ms 12.34.170.33 7 0 ms 0 ms 0 ms 12.34.170.2 Trace complete.</pre>			

FIGURE 39

WG4 PC → EXTERIOR PC

WG4 PC			
Physical	Config	Desktop	F
Command Prompt			
<pre>Cisco Packet Tracer PC Command Line 1.0 C:\>tracert 100.100.100.2 Tracing route to 100.100.100.2 over a maximum of 30 hops: 1 0 ms 0 ms 0 ms 12.34.170.17 2 0 ms 0 ms 0 ms 12.34.170.35 3 0 ms 0 ms 0 ms 12.34.170.50 4 0 ms 0 ms 12 ms 12.34.168.2 5 0 ms 0 ms 0 ms 200.200.200.2 6 0 ms 0 ms 0 ms 100.100.100.2 Trace complete.</pre>			

FIGURE 40

7.4 Investigation and Results

5.1 & 6.1 ROUTING TABLE COMPARISONS

The show ip route command was used to identify routes learned by RIPv2, OSPF, and EIGRP including metrics and administrative preferences.

RIPv2 displayed routes marked with "R", using hop count as its only metric [2], with a maximum of 15 hops and an administrative distance of 120. This made it easily configurable but limited and ill-suited to a dynamic or complex network. It is illustrated in tables by Figures 6, 7, and 8 with no cost or bandwidth awareness.

OSPF used cost-based metrics computed from the formula $\text{Cost} = \text{Reference Bandwidth} / \text{Interface Bandwidth}$. The intra-area routes were designated with an "O" and the inter-area routes were designated with an "O IA" which can be seen in Figures 19, 20, and 21. OSPF, which had an administrative distance of 110, provided greater route trust than RIP, as well as more precise path selection through a heterogeneous set of mixed-speed links.

For EIGRP, the metric was composite, based on bandwidth and delay, and could optionally include reliability and load. Routes appeared as either "D" or "D*", referring to EIGRP-learned routes, as shown in Figures 32, 33, and 34. This reflected route propagation via the DUAL algorithm. EIGRP had the lowest administrative distance of 90 and, therefore, was the most trusted by default; it also combined quick convergence with adaptive metric handling.

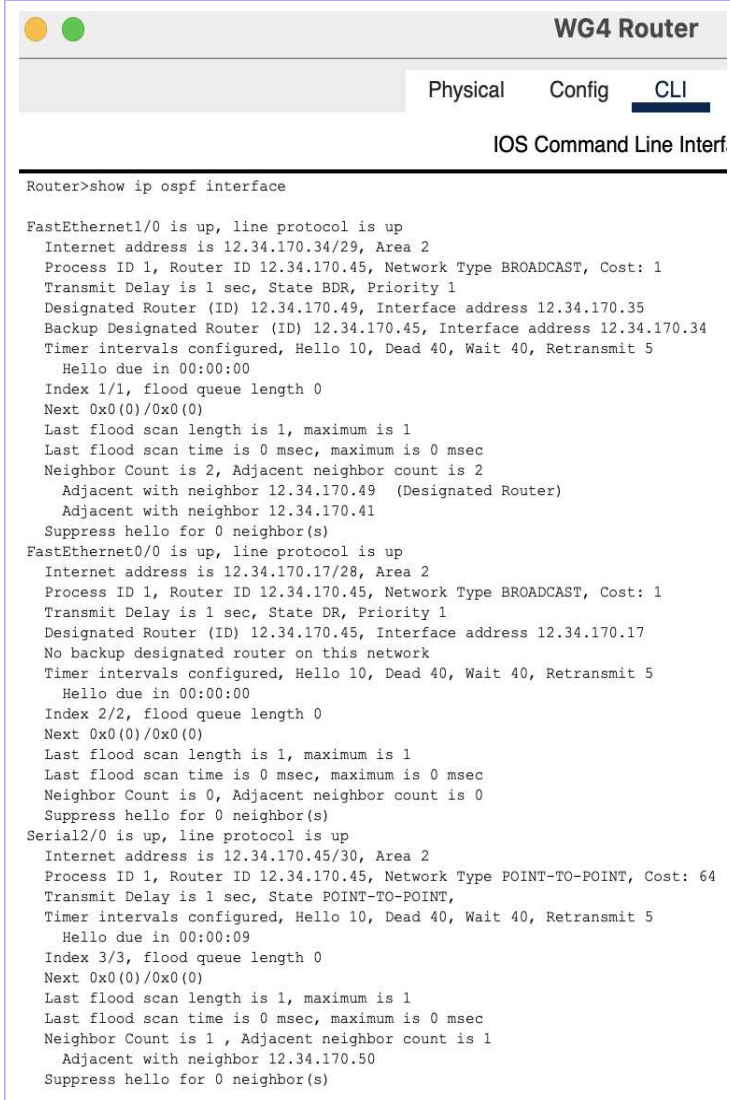
In short, RIPv2 was only adequate for static environments. OSPF displayed scalable routing and cost-awareness in structured networks. EIGRP was most successful overall for its fast convergence and intelligent use of metrics, making it the most sophisticated of the three when in the presence of Cisco support.

5.2 OSPF BANDWIDTH IMPACT ANALYSIS

By default, OSPF calculates its link costs using the following formula: "Cost = Reference Bandwidth / Interface Bandwidth" [8]. Therefore, the reference bandwidth is set to 100 Mbps, which results in all Fast Ethernet interfaces (100 Mbps) being given a cost of 1. When data rates on interfaces start to increase (e.g. Gigabit Ethernet), this becomes a limitation when all higher-speed links are treated the same. Before modification, WG4 Router's Fast Ethernet interfaces had a cost of 1 (Figure 44).

The command "auto-cost reference-bandwidth 1000" was applied on all OSPF routers to ensure more accurate route selection based on interface speed; this can be seen in Figure 45 above. This raised the reference bandwidth to 1000 Mbps and enabled the distinction between 100 Mbps (cost = 10) and 1 Gbps links (cost = 1). This change correctly recalculated the cost as 10 (Figure 45), matching with the OSPF metric formula: $(1000 \div 100)$.

FIGURE 44



```

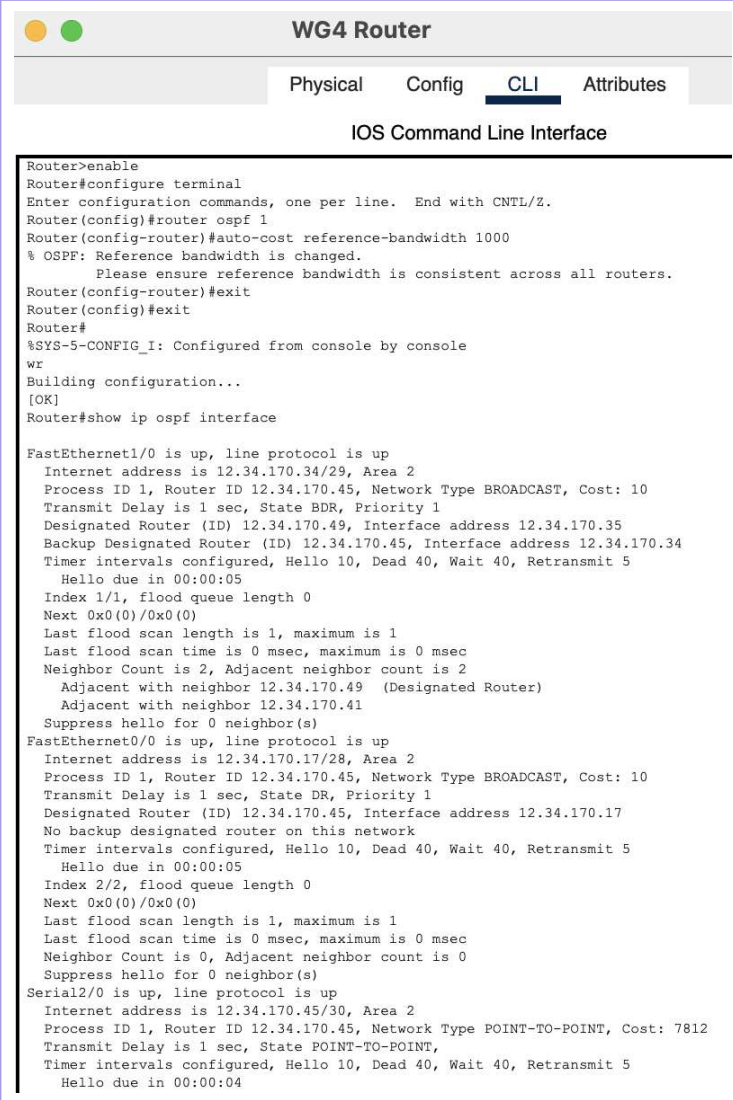
WG4 Router
Physical Config CLI
IOS Command Line Interface

Router>show ip ospf interface

FastEthernet1/0 is up, line protocol is up
  Internet address is 12.34.170.34/29, Area 2
  Process ID 1, Router ID 12.34.170.45, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 12.34.170.49, Interface address 12.34.170.35
  Backup Designated Router (ID) 12.34.170.45, Interface address 12.34.170.34
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:00
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 2, Adjacent neighbor count is 2
    Adjacent with neighbor 12.34.170.49 (Designated Router)
    Adjacent with neighbor 12.34.170.41
  Suppress hello for 0 neighbor(s)
FastEthernet0/0 is up, line protocol is up
  Internet address is 12.34.170.17/28, Area 2
  Process ID 1, Router ID 12.34.170.45, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 12.34.170.45, Interface address 12.34.170.17
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:00
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial2/0 is up, line protocol is up
  Internet address is 12.34.170.45/30, Area 2
  Process ID 1, Router ID 12.34.170.45, Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:09
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 12.34.170.50
  Suppress hello for 0 neighbor(s)

```

FIGURE 45



```

WG4 Router
Physical Config CLI Attributes
IOS Command Line Interface

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
Please ensure reference bandwidth is consistent across all routers.
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
vr
Building configuration...
[OK]
Router#show ip ospf interface

FastEthernet1/0 is up, line protocol is up
  Internet address is 12.34.170.34/29, Area 2
  Process ID 1, Router ID 12.34.170.45, Network Type BROADCAST, Cost: 10
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 12.34.170.49, Interface address 12.34.170.35
  Backup Designated Router (ID) 12.34.170.45, Interface address 12.34.170.34
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:05
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 2, Adjacent neighbor count is 2
    Adjacent with neighbor 12.34.170.49 (Designated Router)
    Adjacent with neighbor 12.34.170.41
  Suppress hello for 0 neighbor(s)
FastEthernet0/0 is up, line protocol is up
  Internet address is 12.34.170.17/28, Area 2
  Process ID 1, Router ID 12.34.170.45, Network Type BROADCAST, Cost: 10
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 12.34.170.45, Interface address 12.34.170.17
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:05
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial2/0 is up, line protocol is up
  Internet address is 12.34.170.45/30, Area 2
  Process ID 1, Router ID 12.34.170.45, Network Type POINT-TO-POINT, Cost: 7812
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:04

```

Observations:

- As shown in WG4 Router's output, FastEthernet interfaces were recalculated with a cost of 10 (Figure 45).
- If any Gigabit interfaces existed, they would show a cost of 1.
- OSPF routing decisions now prioritize faster interfaces where available, allowing more intelligent path selection.

Impact:

This change means that faster connections are now more important when calculating routes in OSPF. Because of this, the SPF algorithm uses better factors, which helps routing work well across the system. The overall performance also improves in places with different connection speeds. This shows how adaptable and broad OSPF is. On the other hand, RIPv2 does not take bandwidth into account when it calculates routes.

5.3 CONVERGENCE TIME ANALYSIS

To evaluate the routing protocols' responsiveness to failure within the topology, a simulated link shutdown (SLS) was applied by disabling the connection between WG2 and ABR1 and the connection between FE Switch 1 and ABR1 as shown in Figure 15. This forced each protocol to reroute the traffic through an alternative path:

- WG2 → FE Switch 1 → WG1 Router → GFE Switch → GFE Router → ABR1

Traceroute results from WG2 PC to the external server (100.100.100.2) were captured before and after the failure for each protocol.

RIP v2

- Before SLS (Figure 12): WG2 used a 6-hop route: 12.34.169.17 → .35 → .18 → .2 → 200.200.200.2 → 100.100.100.2.
- After SLS (Figure 41): RIPv2 recalculated a longer path (8 hops), now going through .17 → .33 → .50 → .18 → .2 → 200.200.200.2 → 100.100.100.2.
- Observation: Delay in route recalculation was observed due to RIP's reliance on periodic updates (~30 seconds). Packets eventually rerouted successfully, but with a brief interruption.

OSPF

- Before SLS (Figure 25): WG2 followed a 6-hop route via .17 → .35 → .18 → .2 → 200.200.200.2 → 100.100.100.2.
- After SLS (Figure 43): Traceroute showed an 8-hop reroute through .17 → .33 → .50 → .18 → .2 → 200.200.200.2 → 100.100.100.2.
- Observation: OSPF responded almost instantly using LSAs and the SPF algorithm, recalculating optimal routes with minimal disruption.

EIGRP (6.1)

- Before SLS (Figure 38): Standard 6-hop route through .17 → .43 → .50 → .18 → .2 → 200.200.200.2 → 100.100.100.2.
- After SLS (Figure 42): Alternative 8-hop path engaged rapidly without manual intervention.
- Observation: EIGRP demonstrated quick convergence using its DUAL algorithm. The reroute was instant and seamless.

These results validate that OSPF and EIGRP are significantly more responsive to topology changes than RIPv2, making them better suited for large or dynamic networks where uptime is critical.

SHUTDOWN TEST SUMMARY

COMMANDS

WG2 Router

```
conf t
interface
GigabitEthernet0/6
shutdown
```

FE Switch 1

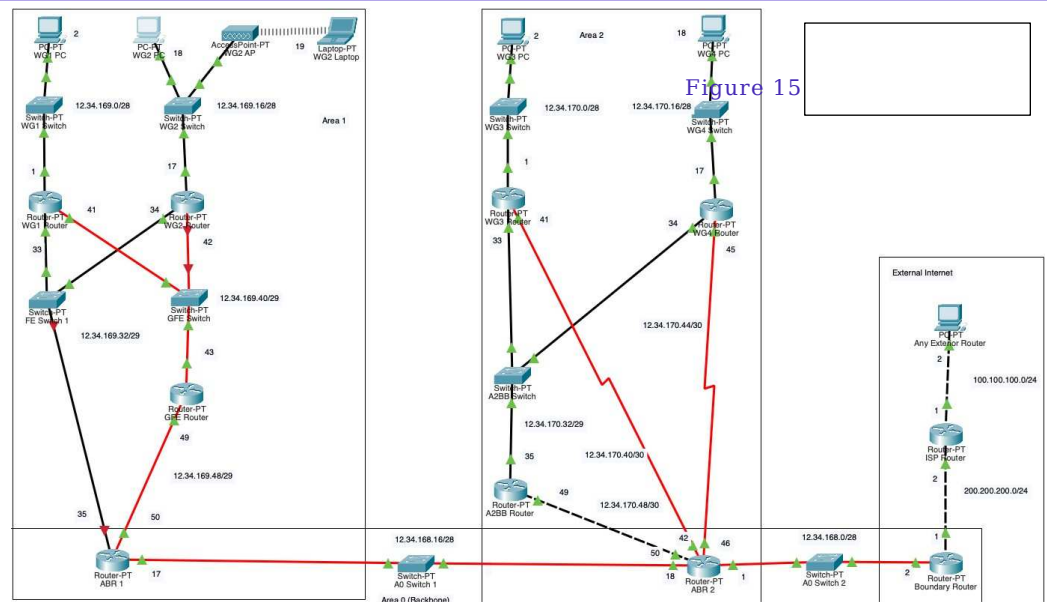
```
conf t
interface
FastEthernet2/1
shutdown
```

Tracert

```
tracert 100.100.100.2
```

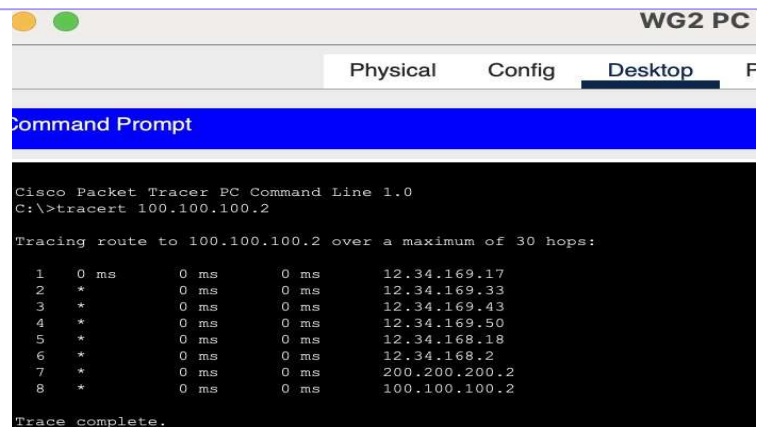
DETAILS

SHUTDOWN CONFIRMATION: WG2 ROUTER + FE SWITCH 1



TRACERT (from WG2 PC)

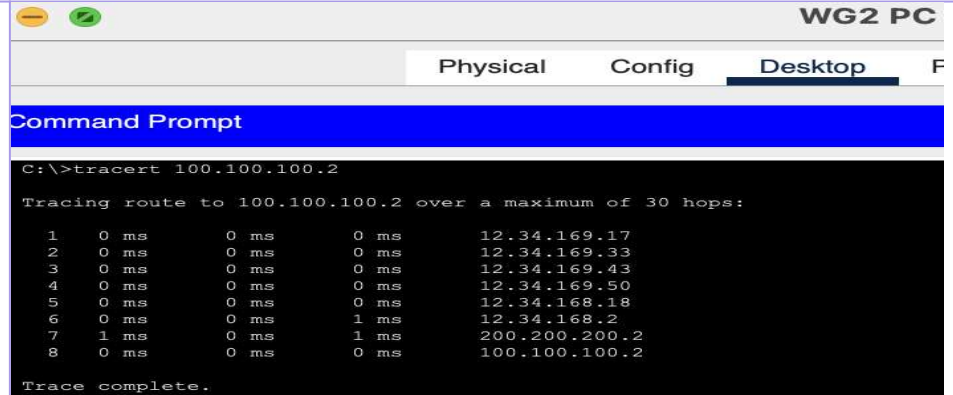
FIGURE 41:
RIPv2



SHUTDOWN TEST SUMMARY

COMMANDS

DETAILS

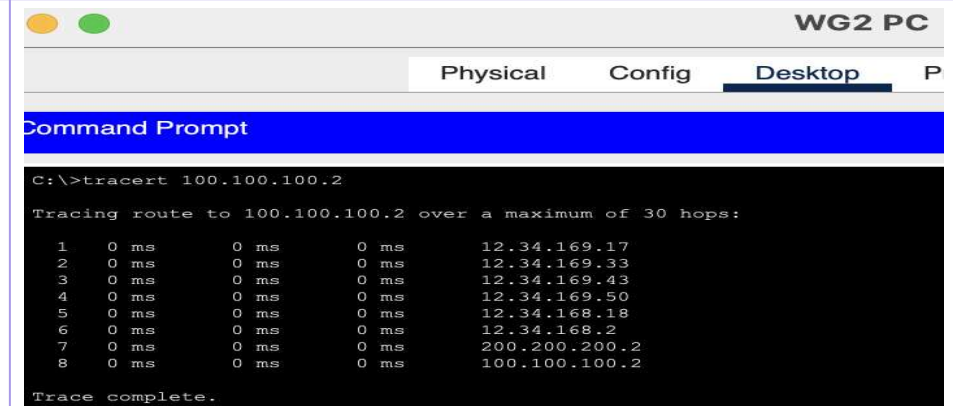
FIGURE 42:
EIGRP

```
WG2 PC
Physical Config Desktop F
Command Prompt
C:\>tracert 100.100.100.2

Tracing route to 100.100.100.2 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    12.34.169.17
  2  0 ms    0 ms    0 ms    12.34.169.33
  3  0 ms    0 ms    0 ms    12.34.169.43
  4  0 ms    0 ms    0 ms    12.34.169.50
  5  0 ms    0 ms    0 ms    12.34.168.18
  6  0 ms    0 ms    1 ms    12.34.168.2
  7  1 ms    0 ms    1 ms    200.200.200.2
  8  0 ms    0 ms    0 ms    100.100.100.2

Trace complete.
```

FIGURE 43:
OSPF

```
WG2 PC
Physical Config Desktop P
Command Prompt
C:\>tracert 100.100.100.2

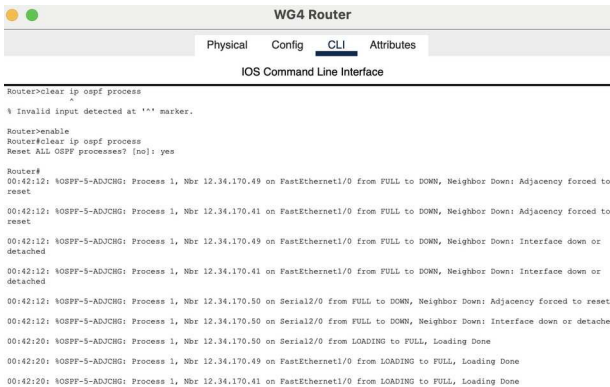
Tracing route to 100.100.100.2 over a maximum of 30 hops:

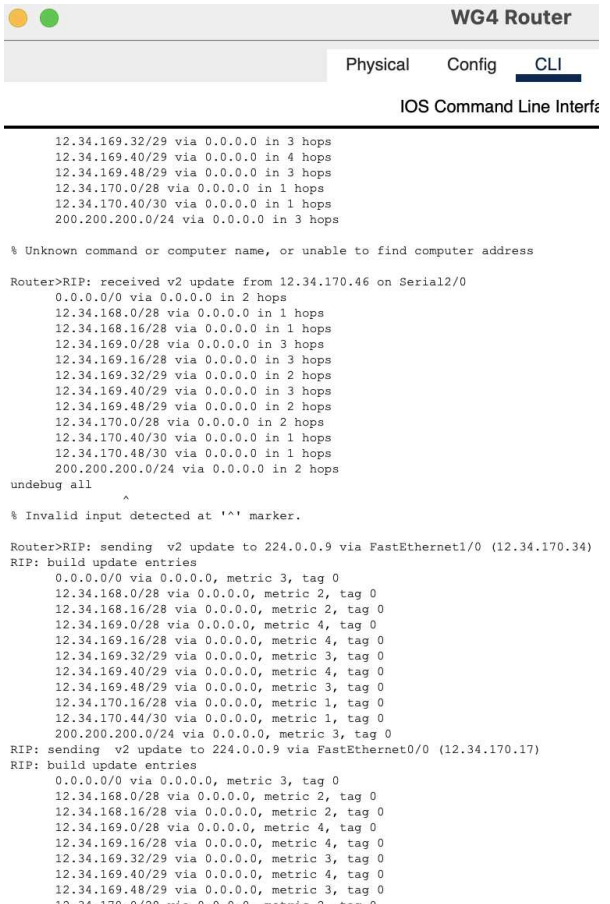
  1  0 ms    0 ms    0 ms    12.34.169.17
  2  0 ms    0 ms    0 ms    12.34.169.33
  3  0 ms    0 ms    0 ms    12.34.169.43
  4  0 ms    0 ms    0 ms    12.34.169.50
  5  0 ms    0 ms    0 ms    12.34.168.18
  6  0 ms    0 ms    0 ms    12.34.168.2
  7  0 ms    0 ms    0 ms    200.200.200.2
  8  0 ms    0 ms    0 ms    100.100.100.2

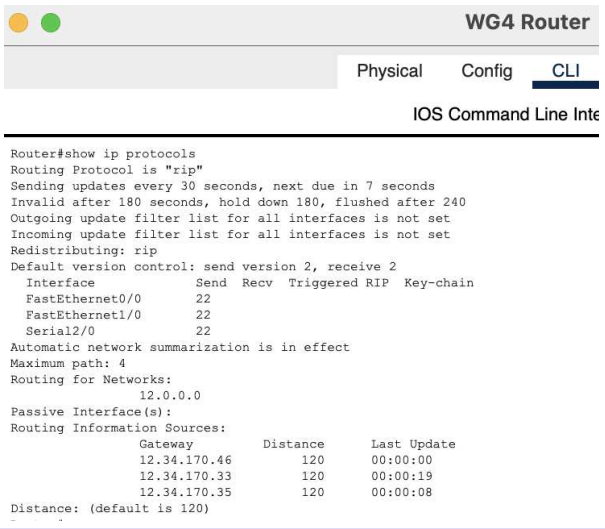
Trace complete.
```

5.4 EXTRA COMMAND RESEARCH

OSPF																																																																																																													
COMMAND	FIGURE 46-48	OBSERVATION																																																																																																											
SHOW IP OSPF NEIGHBOR [10]	WG4 Router PhysicalConfigCLIAttribu IOS Command Line Interface Router>show ip ospf neighbor <table><thead><tr><th>Neighbor ID</th><th>Pri</th><th>State</th><th>Dead Time</th><th>Address</th><th>Interface</th></tr></thead><tbody><tr><td>12.34.170.49</td><td>1</td><td>FULL/DR</td><td>00:00:37</td><td>12.34.170.35</td><td>FastEthernet1/0</td></tr><tr><td>12.34.170.41</td><td>1</td><td>FULL/DROTHER</td><td>00:00:37</td><td>12.34.170.33</td><td>FastEthernet1/0</td></tr><tr><td>12.34.170.50</td><td>0</td><td>FULL/ -</td><td>00:00:37</td><td>12.34.170.46</td><td>Serial2/0</td></tr></tbody></table>	Neighbor ID	Pri	State	Dead Time	Address	Interface	12.34.170.49	1	FULL/DR	00:00:37	12.34.170.35	FastEthernet1/0	12.34.170.41	1	FULL/DROTHER	00:00:37	12.34.170.33	FastEthernet1/0	12.34.170.50	0	FULL/ -	00:00:37	12.34.170.46	Serial2/0	Displays OSPF neighbor states (e.g., FULL/DR, DROTHER), priorities, and links.																																																																																			
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SHOW IP OSPF DATABASE [10]	WG4 Router PhysicalConfigCLI IOS Command Line Int Router>show ip ospf database OSPF Router with ID (12.34.170.45) (Process ID 1) Router Link States (Area 2) <table><thead><tr><th>Link ID</th><th>ADV Router</th><th>Age</th><th>Seq#</th><th>Checksum</th><th>Link cour</th></tr></thead><tbody><tr><td>12.34.170.45</td><td>12.34.170.45</td><td>547</td><td>0x8000000a</td><td>0x00983d</td><td>4</td></tr><tr><td>12.34.170.50</td><td>12.34.170.50</td><td>636</td><td>0x80000008</td><td>0x00ab26</td><td>5</td></tr><tr><td>12.34.170.49</td><td>12.34.170.49</td><td>631</td><td>0x80000006</td><td>0x00d846</td><td>2</td></tr><tr><td>12.34.170.41</td><td>12.34.170.41</td><td>631</td><td>0x80000007</td><td>0x00c30d</td><td>4</td></tr></tbody></table> Net Link States (Area 2) <table><thead><tr><th>Link ID</th><th>ADV Router</th><th>Age</th><th>Seq#</th><th>Checksum</th></tr></thead><tbody><tr><td>12.34.170.50</td><td>12.34.170.50</td><td>636</td><td>0x80000002</td><td>0x004be4</td></tr><tr><td>12.34.170.35</td><td>12.34.170.49</td><td>631</td><td>0x80000003</td><td>0x00f449</td></tr></tbody></table> Summary Net Link States (Area 2) <table><thead><tr><th>Link ID</th><th>ADV Router</th><th>Age</th><th>Seq#</th><th>Checksum</th></tr></thead><tbody><tr><td>12.34.168.0</td><td>12.34.170.50</td><td>626</td><td>0x80000009</td><td>0x005e25</td></tr><tr><td>12.34.168.16</td><td>12.34.170.50</td><td>626</td><td>0x8000000b</td><td>0x00b9b7</td></tr><tr><td>12.34.169.48</td><td>12.34.170.50</td><td>611</td><td>0x8000000c</td><td>0x00a5a0</td></tr><tr><td>12.34.169.32</td><td>12.34.170.50</td><td>611</td><td>0x8000000d</td><td>0x004411</td></tr><tr><td>12.34.169.40</td><td>12.34.170.50</td><td>611</td><td>0x8000000e</td><td>0x00fb4f</td></tr><tr><td>12.34.169.0</td><td>12.34.170.50</td><td>611</td><td>0x8000000f</td><td>0x005b1f</td></tr><tr><td>12.34.169.16</td><td>12.34.170.50</td><td>601</td><td>0x80000010</td><td>0x00b8b0</td></tr></tbody></table> Summary ASB Link States (Area 2) <table><thead><tr><th>Link ID</th><th>ADV Router</th><th>Age</th><th>Seq#</th><th>Checksum</th></tr></thead><tbody><tr><td>200.200.200.1</td><td>12.34.170.50</td><td>626</td><td>0x8000000a</td><td>0x00d815</td></tr></tbody></table> Type-5 AS External Link States <table><thead><tr><th>Link ID</th><th>ADV Router</th><th>Age</th><th>Seq#</th><th>Checksum</th><th>Tag</th></tr></thead><tbody><tr><td>0.0.0.0</td><td>200.200.200.1</td><td>675</td><td>0x80000002</td><td>0x004234</td><td>1</td></tr></tbody></table>	Link ID	ADV Router	Age	Seq#	Checksum	Link cour	12.34.170.45	12.34.170.45	547	0x8000000a	0x00983d	4	12.34.170.50	12.34.170.50	636	0x80000008	0x00ab26	5	12.34.170.49	12.34.170.49	631	0x80000006	0x00d846	2	12.34.170.41	12.34.170.41	631	0x80000007	0x00c30d	4	Link ID	ADV Router	Age	Seq#	Checksum	12.34.170.50	12.34.170.50	636	0x80000002	0x004be4	12.34.170.35	12.34.170.49	631	0x80000003	0x00f449	Link ID	ADV Router	Age	Seq#	Checksum	12.34.168.0	12.34.170.50	626	0x80000009	0x005e25	12.34.168.16	12.34.170.50	626	0x8000000b	0x00b9b7	12.34.169.48	12.34.170.50	611	0x8000000c	0x00a5a0	12.34.169.32	12.34.170.50	611	0x8000000d	0x004411	12.34.169.40	12.34.170.50	611	0x8000000e	0x00fb4f	12.34.169.0	12.34.170.50	611	0x8000000f	0x005b1f	12.34.169.16	12.34.170.50	601	0x80000010	0x00b8b0	Link ID	ADV Router	Age	Seq#	Checksum	200.200.200.1	12.34.170.50	626	0x8000000a	0x00d815	Link ID	ADV Router	Age	Seq#	Checksum	Tag	0.0.0.0	200.200.200.1	675	0x80000002	0x004234	1	Reveals link-state database structure, showing LSAs and router link states.
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OSPF		
COMMAND	FIGURE 46-48	OBSERVATION
CLEAR IP OSPF PROCESS [12]		Resets OSPF adjacencies and forces recomputation of routes; useful for testing.

RIP V2		
COMMAND	FIGURE 49-51	OBSERVATION
DEBUG IP RIP [11]		Real-time monitoring of RIP updates (send/receive). Useful to visually confirm periodic update intervals and see if a router is advertising or receiving incorrect networks. Note: make sure to only use in lab set ups, causes console overloading. Hence why the typing of the command is not present within Figure

RIP V2		
COMMAND	FIGURE 49-51	OBSERVATION
SHOW IP PROTOCOLS [11]	 <pre> WG4 Router Physical Config CLI IOS Command Line Inte Router#show ip protocols Routing Protocol is "rip" Sending updates every 30 seconds, next due in 7 seconds Invalid after 180 seconds, hold down 180, flushed after 240 Outgoing update filter list for all interfaces is not set Incoming update filter list for all interfaces is not set Redistributing: rip Default version control: send version 2, receive 2 Interface Send Recv Triggered RIP Key-chain FastEthernet0/0 22 FastEthernet1/0 22 Serial2/0 22 Automatic network summarization is in effect Maximum path: 4 Routing for Networks: 12.0.0.0 Passive Interface(s): Routing Information Sources: Gateway Distance Last Update 12.34.170.46 120 00:00:00 12.34.170.33 120 00:00:19 12.34.170.35 120 00:00:08 Distance: (default is 120) - - </pre>	Reveals link-state database structure, showing LSAs and router link states.
CLEAR IP ROUTE * [13]	 <pre> WG4 Router Physical Config CLI IOS Command Line Inte Router#clear ip route * Router#exit </pre>	Resets all RIPv2 adjacencies and forces recomputation of routes; useful for testing. (* = all)

Topology and Addressing Scheme Critique

Although the topology issued for this project met its intended goals, after further review, testing and analysis of the network, there are areas which could be significantly improved through better structural planning and IP management. Roles and functions were not visually separated in the design. Because the devices were not organised by layer or function, the topology was tougher to comprehend and less maintainable. A better-organised layout that logically segments LANs, distribution layers, and core devices would benefit scalability and troubleshooting drastically.

Functionally speaking, there was no inherent redundancy for important devices. ABR1 and WG1 routers acted as single points of failure. Because of this, the network is fragile and unsuited for practical deployments where uptime is essential. An addition of backup links and alternate paths which can be navigated through via RIP or other protocols in-case of emergency, would greatly improve resilience within the topology.

The addressing scheme made use of public IP ranges e.g 12.34.169.0/24 for internal networks. To prevent possible conflicts, private address ranges like 10.0.0.0/8 or 192.168.0.0/16 should be used in its place. Although the VLSM approach was successful, it could still be further optimised. This could be done by an assignment of /30 subnets for point-to-point links and /24 subnets for LANs which would have conserved address space as well as improving overall clarity.

Finally, OSPF router IDs were assigned using physical interfaces. This can lead to issues if interfaces go down or change. Allocating loopback interfaces for OSPF router IDs is a standard best practice and would ensure consistent identification regardless of physical changes. [9]

Summary

To summarise, this report met the CI5210 brief by configuring and comparing RIPv2 and OSPF using Cisco Packet Tracer. The theory of all three protocols was briefly explained before they were tested for route learning, communication, convergence, and failure response. Verification was carried out using ping, tracert, and show ip route, both with the fully functioning topology as well as a simulated link shutdown test to see how each protocol would handle it, showing resilience level for comparison. Further experiments compared routing tables, route metrics, evaluated OSPF's performance under adjusted reference bandwidth as well as additional commands which were explored for deeper protocol insights.

For extra credit, two extra stages were completed, EIGRP was implemented to study hybrid routing, and a design critique was written to identify areas of improvement, including IP addressing, redundancy, and the use of loopbacks. The report ends with a clear analysis showcasing a practical understanding of how routing protocols behave and how they apply in real-world networks as well as showcasing how they are configured in detail.

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Appendix A

5. Investigations

1. For a selection of routers, compare the routing tables produced by RIP and OSPF respectively. What differences do you notice between them? In particular, look at the numbers of routes in the tables and the route metrics. Can you explain what you see?
2. For the OSPF system, increase the reference bandwidth to 1000Mbit/s and observe the effect upon the routing tables. Again, can you explain what you see?
3. Compare the speeds at which OSPF or RIPv2 respond to changes in the topology. Can you explain the difference?
4. Research the different commands associated with RIPv2 and OSPF. Experiment with these commands and record your findings. Can you explain your observations?

6. Suggestions for Extra Credit

1. Investigate a third routing protocol (e.g. EIGRP) and compare its behaviour with that of RIPv2 and OSPF.
2. Critique the internetwork topology and addressing scheme. Could it have been better designed? If so, what would you have done differently? Give reasons for your decisions.

7. Deliverable

Present your findings as a technical report. (No fixed word count, but there are no extra marks for unnecessary length. Approximately 1000-2000 words are recommended.) Your report should contain:

1. A very brief introductory section explaining what a routing protocol is, and setting out the aims of the investigation.
2. A theory section, explaining the concepts behind the protocols to be investigated (e.g. link state vs. distance vector, how route metrics are computed etc.)
3. A brief explanation of how the routing protocols were configured, the commands used, and evidence that correct operation has been verified. (You need not include a separate screenshot for every command entered and every ping test performed – a few well-selected examples will be fine.)
4. A description of your investigations from Section 5 (and possibly 6), including screenshots and reasoned commentary on your findings.
5. A brief conclusion, summing up the report and what you have learned.
6. References, presented in a recognised academic style. These should all be cited inline in the text.

Source: <https://canvas.kingston.ac.uk/courses/29971/assignments/183478>